Reading report
PhD candidate: Maxime Deforche
Title of the doctoral dissertation: Management and Analysis of Short Texts: A Graph-Based Approach
Member of the examination board: Joachim Nielandt

General

The thesis is particularly well-structured and was a pleasure to read. Used academic English language (cannot comment on the Greek) is comprehensible and well-crafted. The way each of the chapters is put together and builds on top of what was presented previously is well done: the thesis is clearly a coherent whole.
Technicalities are rigorously explained, with numerous examples where necessary, making the thesis quite digestible for readers that are no domain experts. Results, too, are interpreted expansively. It is clear that you understand the principles at work, as unexpected deviations are explained well and technical reasons for specific results are clarified when necessary.
Additionally, as this work is part of an overarching project, this was also presented clearly to a reader that has limited background knowledge. It is good to know this research will be taken up within the scope of an existing project, with a willing community of experts.

Structure

State of the art was sprinkled throughout, for each of the relevant sections. I found it to be more than adequate to paint a picture of what is available, and how the current work could be situated within existing options.
Research questions are clearly stated, referenced to and repeated when necessary. They also form a coherent story on which the whole thesis rests, and they integrate particularly well.
Main contributions, likewise, are clearly described and summarized for the reader.
Illustrations were used sparingly but effectively, and together with some choice result samples they really tied the room together.

Conclusion

Once again, this thesis was a pleasure to read. I am still in shock that I could not spot a single blatant typo. I look forward to the presentation!
Below, I leave some general feedback on the text. It could help you clarify or fix some parts, though obviously feel free to disregard suggestions if they do not make sense to you.

Text feedback

I used page numbers from the pdf, which do not correspond to the internal count. They should be consistent though. Feel free to send me an email for clarification if you'd need some!

- general: line numbers are present (part of the proof, should be present in the final file?)
- general: references to an Equation are done with brackets, whereas references to Sections and other elements are not, i.e., "Equation (6.17)" but "Section 4.5.3". I'd eliminate the brackets if possible, seems cleaner and more consistent.
- p14 - "querying and visualising the graph make patterns" - make > makes
- p35 - "with particular attention on textual data." - on > to

- ~~p37 - "Trained on enormous datasets of human-generated text," - datasets > theft~~
- p39 - "that big data poses" - poses > pose
- p41 - "outputs of such analysis" - analysis > analyses (?)
- p41 - "to support this process" - perhaps replace by "to support knowledge extraction", as it's not directly clear to me at this point what process is referred to
- p42 - "the use of flexible querying techniques, which allow" - i *think* it's "allows", as the subject is "the use"
- p43 - "unlike structured data, which conforms" - conforms > conform
- p46 - footnote two could be typeset a bit prettier (url of dbbe), perhaps by prefixing it with "DBBE homepage: " so that the supertext 2 doesn't stick to the blue mono text?
- p50 - "include POS tags" - first time POS is used I think, add "(part of speech)" ?
- p51 - "In many real-world scenarios, contextual information is only partially available or entirely absent for part of the data." - "for part of the data" seems superfluous/unclear here, feels like you can remove that part of the sentence without losing meaning?
- p52 - "but can include partially similar or structurally related texts" - can partially be removed without losing meaning?
- p69 - "Hierarchical graph-based decompositions instead retain" - the 'instead' reads in a funky way, as you repeat what came before... to me it would read easier as follows: "Hierarchical graph-based decompositions however, do: ..."
- p70 - "The model can, however, serve as the foundation from which features" - the > a
- p80 - "we use a Neo4j graph database system" - a > the (or "a neo4j database")
- p81 - "The examples presented in this section are based on epigrams belonging to three groups (epigram types) in the DBBE" - concepts "epigram type" and "group" are introduced here but were not explained beforehand. Could perhaps be done to better understand how a "type" fits in the hierarchy.
- p120 - "setting the main quantifier parameter" - there was some confusion on my part at this point in the text, as alpha was previously also used. Or is alpha in this context equivalent to the proposed method's alpha cut?
- p130 - "In epigram 22826, the word βλεπιν is used where epigram 21182 uses ιδιν, which are synonymous" - not sure "which are" is correct here as there is no subject. perhaps better phrased as "epigrams x and y respectively use the synonymous words a and b"?
- p133 - It's mentioned that the expert grouping are not perfect, perhaps to be mentioned in the main text too as this only pops up in the conclusion as far as I read. Later on you mention this fact lots of times, but perhaps could be done as well at first introduction of the expert groupings.
    - also see 166: "scores reported here are therefore best read as a comparison against one defensible scholarly interpretation of the corpus, not as an absolute measure of quality."
- p145 - "and its sorting-based semantics, which make it" - "and its sorting-based semantics that make it" (the amount of commas and the which makes for confusion on my part to what the which is actually referring to)

- p150 - "when the values of a field can be represented as sets" - as sets > as a set?
- Figure 4.1: c is not defined yet at the point of reading, caption does not mention what it is. Perhaps worth adding it to avoid having to look it up. s isn't mentioned in the caption either, to be fixed in the same fashion I guess.
- p154 - "none of them has been applied" - i'd personally use "have" instead of "has" here, as it refers to multiple techniques
- p189 - "injective" - perhaps quickly explain what this means, as the word only occurs once in the text
- p210 - "hierarchical geographical structure and assigned satisfaction" - i'd repeat the are ("and are assigned"), disambiguating between the epigrams that assign something, and the epigrams that are assigned something
- while reading your thesis, and the quotes, my mind drifted to the lyrics of a song that features in The Big Lebowski (obligatory cameo in every thesis)... would have been a nice one as well: "I just dropped in to see what condition my condition was in" (https://www.azlyrics.com/lyrics/kennyrogers/justdroppedintoseewhatconditionmyconditionwasin.html). Would be a shoe-in for *confidence* on a *similarity!*
- p230 - "The two are computed from disjoint inputs" - it's not 100% clear to me what "the two" means when reading the text, based on the previous sentence. I assume it's the similarity and the confidence?
- p235 - "as the ground truth" - i'd remove "the"
- p236 - "contextual metadata are incorporated and, when they are incorporated, in the weight vectors" - ... and, if so, in the weight ... (but for sure, if instead of when?)
- p236 - "(ctx for context-aware without confidence, nw for non-weighted, and w for weighted) and the subscript denotes the clustering algorithm (KMedoids, HAC, or FCMdd)" - *ctx* is italic, but KMedoids is not. i.e., the superscripts are shown differently than the subscripts. it would perhaps be a bit clearer here to say "w^ctx for ..." and "w_hac" for ..., that way it visually points back to the weight variable that contains both strategy and cluster super- and subscripts
- p238 - "For FCMdd, the fuzzification exponent q is selected in the same search rather than fixed in advance." - "in the same search" is unclear to me, not sure what you mean here. Which search (same as ...?). Selecting q in each of the stages? ... Could be rephrased I think, for clarification.
- p239 - "the ARI and AMI measures address this limitation."- ARI (or AMI) is first mentioned here, but it's not defined yet. Perhaps clarify that they will be explained below in the following paragraphs, especially as their titles don't contain the abbreviations either.
- 247 - "relationships even where the metadata" - where>when?
- p251 - "where the distinct groups of epigrams are the clusters and the edges connect" > put a comma before "and the edges", at first glance "the clusters and the edges" is the object of "are" to me